

LP Solvable formulations for OWA and Choquet models

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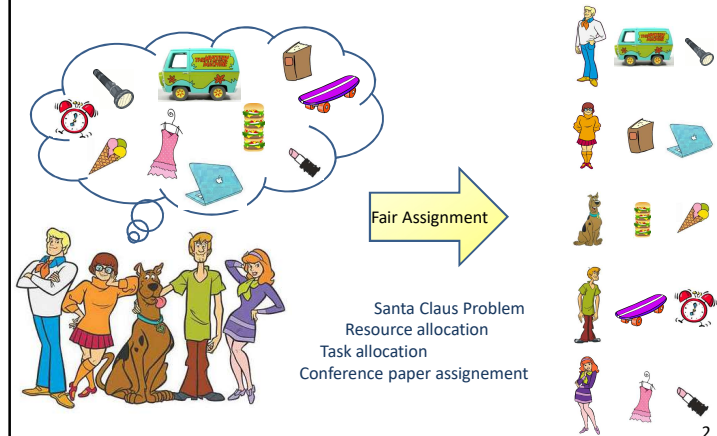


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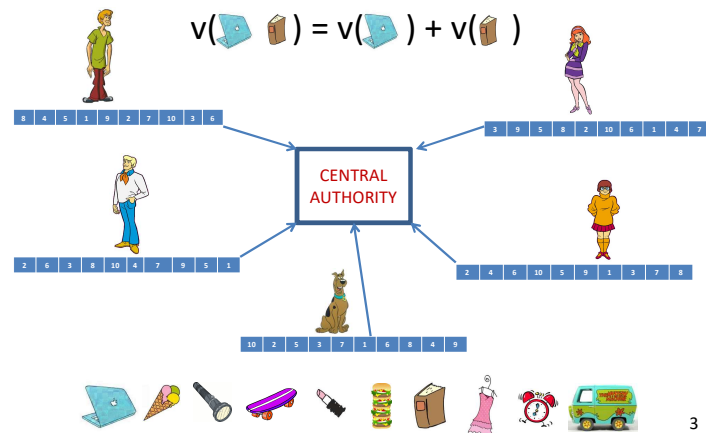
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Multiagent Fair Assignment Problems



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Additivity and communication costs



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Fairness in the Santa Claus problem

	Box	Ice cream	Skateboard	Book	Dress
1	5	8	(4)	9	7
2	1	(3)	2	7	8
3	(3)	9	2	9	5
4	10	1	3	(3)	4
5	5	1	7	7	(3)

Costs or disutilities

1	1	2	3	7
$\Sigma=14$	$\max=7$			

3	3	4	3	3
$\Sigma=16$	$\max=4$			

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Fairness in paper assignment problems

2 reviewers per paper

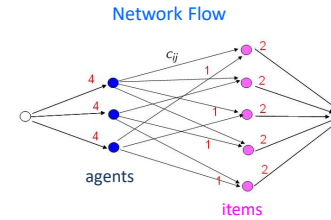
u_{ij}	Paper 1	Paper 2	Paper 3	Paper 4	Paper 5
Reviewer 1	3	3	4	3	4
Reviewer 2	3	4	4	2	3
Reviewer 3	1	2	3	2	3

Solution 1 $1 \leftarrow \{1, 3, 4, 5\}$ $2 \leftarrow \{1, 2, 3\}$ $3 \leftarrow \{2, 4, 5\}$
 $x = (14, 11, 7)$ $\Sigma=32$ $\min = 7$

Solution 2 $1 \leftarrow \{1, 4, 5\}$ $2 \leftarrow \{1, 2, 3\}$ $3 \leftarrow \{2, 3, 4, 5\}$
 $x = (10, 11, 10)$ $\Sigma=31$ $\min = 10$

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Modeling assignment problems



MO Linear Programming

$$\begin{aligned} \max x_i &= \sum_{j=1}^m u_{ij} z_{ij} \quad i = 1, \dots, n \\ \text{s.t. } &\begin{cases} l_i \leq \sum_{j=1}^m z_{ij} \leq u_i & i = 1, \dots, n \\ l'_j \leq \sum_{i=1}^n z_{ij} \leq u'_j & j = 1, \dots, m \\ z_{ij} \in \{0, 1\} & \forall i, \forall j \end{cases} \end{aligned}$$

Min Cost Max flow → does not guarantee that optimal solutions will be fair!

Bottleneck optimization → (min max cost, max min utility) NP-hard [Bouveret et al., 05]
 focuses on the least satisfied agent... ex for costs : (9,9,9,9,9) < (10,1,1,1,1)

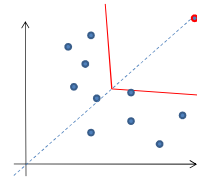
Multiobjective optimization → intractable (too much Pareto-optimal solutions)

What else? → Compromise Search in MCDM, Fairness in Social Choice

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Compromise Search in MCDM

$$\begin{aligned} \max \min_{i=1 \dots n} x_i + \varepsilon \sum_{i=1}^n x_i \\ \text{s.t. } &\begin{cases} x_i = \sum_{j=1}^m u_{ij} z_{ij} & i = 1, \dots, n \\ l_i \leq \sum_{j=1}^m z_{ij} \leq u_i & i = 1, \dots, n \\ l'_j \leq \sum_{i=1}^n z_{ij} \leq u'_j & j = 1, \dots, m \\ z_{ij} \in \{0, 1\} & \forall i, \forall j \end{cases} \end{aligned}$$



$$\max z + \varepsilon \sum_{i=1}^n \sum_{j=1}^m u_{ij} z_{ij}$$

Linearization

$$\text{s.t. } \begin{cases} z \leq \sum_{j=1}^m u_{ij} z_{ij} & i = 1, \dots, n \\ l_i \leq \sum_{j=1}^m z_{ij} \leq u_i & i = 1, \dots, n \\ l'_j \leq \sum_{i=1}^n z_{ij} \leq u'_j & j = 1, \dots, m \\ z_{ij} \in \{0, 1\} & \forall i, \forall j \end{cases}$$

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Fair OWA-optimization for min-cost assignment problems

$$\text{Min } W(x) = \sum_{k=1}^n w_k x_{(k)} \quad w_1 > w_2 > \dots, w_n > 0$$

$$\text{s.t. } \begin{cases} x_i = \sum_{j=1}^m c_{ij} z_{ij} & i = 1, \dots, n \\ l'_j \leq \sum_{i=1}^n z_{ij} \leq u'_j & j = 1, \dots, m \\ l_i \leq \sum_{j=1}^m z_{ij} \leq u_i & i = 1, \dots, n \\ z_{ij} \in \{0, 1\} & \forall i, \forall j \end{cases}$$

Proposition

As soon as $m > 1$, finding an OWA optimal assignment is NP-hard

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Équité et principe de transfert

L'équité peut être exprimée à l'aide du principe de transfert :

PRINCIPLE DE TRANSFERT

Soit $x \in \mathbb{R}_+^n$ tel que $x_i > x_j$ pour $i, j \in N$. Alors, pour tout ε tel que $0 < \varepsilon < x_i - x_j$, $x - \varepsilon e_i + \varepsilon e_j \succ x$ où e_i (resp. e_j) est le vecteur dont la $i^{\text{ème}}$ (resp. $j^{\text{ème}}$) composante vaut 1, toutes les autres étant nulles.

Exemple : $x = (10, 10)$ est préféré au vecteur $y = (14, 6)$ car il existe un transfert de taille 4 permettant de passer de y à x .

Remarque : Le principe de transfert fonctionne à moyenne constante. Il ne permet donc pas d'établir de préférence entre deux vecteurs dont la performance moyenne n'est pas la même.

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Dominance de Lorenz

Pour comparer les vecteurs $(11, 11)$ et $(12, 9)$, on peut remarquer que $(11, 11) \succ_P (11, 10)$ et que $(11, 10) = (12 - 1, 9 + 1)$. On a donc respectivement $(11, 11) \succ (11, 10)$ (principe de Pareto) et $(11, 10) \succ (12, 9)$ (principe de transfert) d'où, par transitivité, $(11, 11) \succ (12, 9)$.

VECTEUR DE LORENZ

Pour tout $x \in \mathbb{R}_+^n$, le *vecteur de Lorenz* associé à x est défini par :

$$L(x) = (x_{\sigma(1)}, x_{\sigma(1)} + x_{\sigma(2)}, \dots, x_{\sigma(1)} + x_{\sigma(2)} + \dots + x_{\sigma(n)})$$

où σ représente la permutation qui réordonne les composantes de x par ordre croissant.

DOMINANCE DE LORENZ GÉNÉRALISÉE

La dominance de Lorenz généralisée est un ordre partiel sur $\mathbb{R}_+^n$ défini par :

$$\forall x, y \in \mathbb{R}_+^n, x \succ_L y \iff L(x) \succ_P L(y)$$

Équité et dominance de Lorenz

THÉORÈME

Pour toute paire de vecteurs $x, y \in \mathbb{R}_+^n$, si $x \succ_P y$, ou si x se déduit de y par un transfert de Pigou-Dalton, alors $x \succ_L y$. Inversement si $x \succ_L y$, alors il existe une séquence de transferts de Pigou-Dalton et/ou d'améliorations au sens de Pareto qui permettent de transformer y en x .

Exemple : $L(11, 10) = (10, 21)$ tandis que $L(12, 9) = (9, 21)$. On a donc $(11, 10) \succ_L (12, 9)$ puisque $(10, 21) \succ_P (9, 21)$.

Remarque : $x \succ_P y \Rightarrow x \succ_L y$ ce qui montre que la dominance de Lorenz est potentiellement plus discriminante que celle de Pareto.

La dominance de Lorenz apparaît donc comme un raffinement naturel de la dominance de Pareto permettant de privilégier les solutions Pareto-optimales qui ont des profils de performances équilibrés.

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Dominance de Lorenz et OWA

En remarquant que : $x_{\sigma(i)} = L_i(x) - L_{i-1}(x)$ pour $i > 1$ il vient :

PROPOSITION

$$OWA(x) = \sum_{i=1}^{n-1} (w_i - w_{i+1}) L_i(x) + w_n L_n(x)$$

On constate donc que si les coefficients w_i sont positifs et choisis de manière à décroître strictement lorsque i augmente, OWA est une combinaison linéaire à coefficients strictement positifs des composantes du vecteur de Lorenz. Par conséquent l'ordre partiel défini par :

$$x \succ_{\text{owa}} y \iff OWA(x) > OWA(y)$$

étend la dominance de Lorenz, c'est-à-dire que $x \succ_L y \Rightarrow x \succ_{\text{owa}} y$. Ainsi OWA utilisé avec des poids w_i strictement décroissants est un agrégateur qui permet de favoriser les solutions équilibrées (très utilisé en optimisation équitable).

A MIP formulation for fair minimization

$$y = (y_1, \dots, y_n) \quad y_{(1)} \geq y_{(2)} \dots \geq y_{(n)}$$

$$\text{OWA}(y) = \sum_{k=1}^n w_k y_{(k)} = \sum_{k=1}^n w'_k L_k(y) \quad w' = (w'_1 - w'_2, \dots, w'_{n-1} - w'_n, w'_n) > 0$$

$$L_k(y) = \begin{array}{ll} \text{Max} & \sum_{i=1}^n \alpha_i^k y_i \\ \text{s.t.} & \sum_{i=1}^n \alpha_i^k = k \\ & 0 \leq \alpha_i^k \leq 1 \quad i = 1 \dots n \end{array} \quad \begin{array}{ll} \text{Min} & kr_k + \sum_{i=1}^n b_i^k \\ \text{s.t.} & r_k + b_i^k \geq y_i \\ & b_i^k \geq 0 \quad i = 1 \dots n \end{array} \quad \text{dual}$$

$$\text{Min} \sum_{k=1}^p w'_k \left(kr_k + \sum_{i=1}^n b_i^k \right)$$

$$r_k + b_i^k \geq y_i$$

$$b_i^k \geq 0$$

Ogryczak, 03

$$\text{Min} \sum_{k=1}^n w'_k \left(k \times r_k + \sum_{i=1}^n b_{ik} \right)$$

$$\text{s.t.} \begin{cases} l_j^k \leq \sum_{i=1}^n z_{ij} \leq u_j^k & j = 1, \dots, m \\ l_i \leq \sum_{j=1}^m z_{ij} \leq u_i & i = 1, \dots, n \\ r_k + b_{ik} \geq \sum_{j=1}^m c_{ij} z_{ij} & \forall i, k = 1, \dots, n \\ b_{ik} \geq 0 & \forall i, k \\ z_{ij} \in \{0, 1\} & \forall i, j \end{cases}$$

Final MIP Formulation

► Extends to agents with exogenous rights (OWA → WOWA)

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Another formulation for the general case

Using linear combination of Lorenz components does not work for non-decreasing weights

Another approach : component $y_{(k)}$ can be obtained by math progr as follows

$$(\Pi_{x_k}^{\uparrow}) \quad \begin{array}{ll} \text{max} & y_k \\ \text{s.t.} & \begin{cases} y_k \leq x_i + M b_i^k & \forall i \in N \\ \sum_{i \in N} b_i^k = k - 1 \end{cases} \\ & y_k \geq 0 \\ & b_i^k \in \{0, 1\} \quad \forall i \in N \end{array}$$

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MIP formulation for general OWA optimization

$$\text{max} \sum_{k \in N} w_k y_k$$

$$\text{s.t.} \begin{cases} y_k \leq x_i + M b_i^k & \forall i \in N, \forall k \in N \\ \sum_{i \in N} b_i^k = k - 1 & \forall k \in N \end{cases}$$

$$y_k \geq 0 \quad \forall k \in N$$

$$b_i^k \in \{0, 1\} \quad \forall i \in N, \forall k \in N$$

$$x \in X$$

Drawback : introduction of N^2 boolean variables, M method

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Infinite order Lorenz Dominance

Presentation in the case of minimization : x cost vector

$L_i(x)$ is the sum of the i largest costs in x ex : $L(1, 4, 2, 3) = (4, 7, 9, 10)$

k -th order Lorenz Dominance : $\prec_L^k$

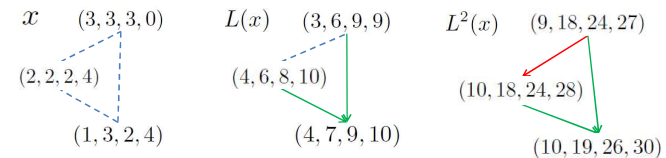
$$L^k(x) = \begin{cases} L(x) & \text{if } k = 1 \\ L(L^{k-1}(x)) & \text{if } k > 1 \end{cases}$$

$$\forall x, y \in \mathbb{R}_+^n, \quad x \prec_L^k y \iff L^k(x) \prec_P L^k(y)$$

Infinite order

$$\prec_L^\infty = \bigcup_{k \geq 1} \prec_L^k$$

($\prec_L^k \subseteq \prec_L^{k+1}$, nested orders)



L-iteration mechanism

i) Progressive refinements ii) preserves Pigou-Dalton principle iii) Decisive?

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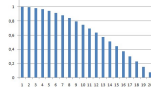
∞ -order Lorenz Dominance

Convergence and decisiveness of the L-iteration mechanism?

A representation theorem for L^∞ -dominance

$$\forall x, y \in \mathbb{R}_+^n, x \prec_L^\infty y \iff W(x) < W(y)$$

$$\text{where } W(x) = \sum_{k=1}^n \sin\left(\frac{(n+1-k)\pi}{2n+1}\right) x_{(k)}$$



- $W(x)$ is an ordered average operation (OWA) with *decreasing weights*, i.e.

Fair Assignment problem

$$\begin{aligned} \text{Min } W(x) &= \sum_{k=1}^n \sin\left(\frac{(n+1-k)\pi}{2n+1}\right) x_{(k)} \\ \text{s.t. } \begin{cases} x_i = \sum_{j=1}^m c_{ij} z_{ij} & i = 1, \dots, n \\ l'_j \leq \sum_{i=1}^n z_{ij} \leq u'_j & j = 1, \dots, m \\ l_i \leq \sum_{j=1}^m z_{ij} \leq u_i & i = 1, \dots, n \\ z_{ij} \in \{0, 1\} & \forall i, \forall j \end{cases} \end{aligned}$$

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The Choquet integral

Capacity

set function $v : 2^N \rightarrow [0, 1]$ such that :

- ▶ $v(\emptyset) = 0, v(N) = 1$
- ▶ $\forall A, B \in 2^N$ such that $A \subseteq B, v(A) \leq v(B)$

Choquet Integral

$$C_v(x) = \sum_{i=1}^n [v(X_{(i)}) - v(X_{(i+1)})] x_{(i)} = \sum_{i=1}^n [x_{(i)} - x_{(i-1)}] v(X_{(i)})$$

where $X_{(i)} = \{j \in S, x_j \geq x_{(i)}\} = \{(i), (i+1), \dots, (n)\}$

Example Utility vector $(x_1, x_2, x_3) = (10, 6, 12)$

$$C_v(10, 6, 12) = 6 v(\{1, 2, 3\}) + (10 - 6) v(\{1, 3\}) + (12 - 10) v(\{3\})$$

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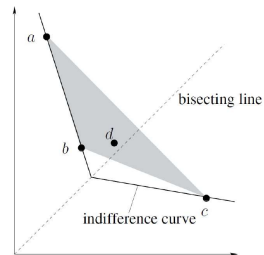
Fairness in the Choquet integral

Preference for interior points

$$\forall x^1, \dots, x^n \in \mathbb{N}^n, \forall \alpha_1, \dots, \alpha_n \geq 0 \text{ s.t. } \sum_{i=1}^n \alpha_i = 1,$$

$$[x^1 \sim x^2 \sim \dots \sim x^n] \Rightarrow \sum_{i=1}^n \alpha_i x^i \succsim x^k, \quad k = 1, \dots, n$$

$(15, 15, 0) \sim (15, 0, 15) \sim (0, 15, 15)$
but $(10, 10, 10)$ is better



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Fairness in the Choquet integral

Preference for interior points

$$\forall x^1, \dots, x^n \in \mathbb{N}^n, \forall \alpha_1, \dots, \alpha_n \geq 0 \text{ s.t. } \sum_{i=1}^n \alpha_i = 1,$$

$$[x^1 \sim x^2 \sim \dots \sim x^n] \Rightarrow \sum_{i=1}^n \alpha_i x^i \succsim x^k, \quad k = 1, \dots, n$$

Proposition [Chateauneuf et Tallon, 02]

Within CEU Theory, preference for interior points is equivalent to choosing v convex

$$v(A \cup B) + v(A \cap B) \geq v(A) + v(B) \text{ for all } A, B \subseteq N$$

Proposition

As soon as $m > 1$, finding an Choquet-optimal assignment for a convex capacity v is NP-hard

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An important notion: the core of a capacity

DUAL CAPACITY

$\bar{v}$ dual capacity of v if: $\forall A \subseteq N, \bar{v}(A) = 1 - v(S \setminus A)$

CORE

- **core of a capacity**: with $\mathcal{L}$ the set of proba. on N
 $\text{core}(v) = \{P \in \mathcal{L}, v(A) \leq P(A) \leq \bar{v}(A)\}$
- v convex $\Rightarrow \text{core}(v) \neq \emptyset$

$$C_v(x) = \min_{\lambda \in \text{core}(v)} \sum_{i=1}^n \lambda(\{i\}) x_i$$

for any convex capacity

(Schmeidler 86)

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Choquet-optimization : linearization 1

$$\begin{aligned} \min \quad & \sum_{i=1}^n \lambda_i x_i \\ \text{s.t.} \quad & \left\{ \begin{array}{l} v(A) \leq \sum_{i \in A} \lambda_i \quad \forall A \subseteq N \\ \lambda_i \geq 0 \quad i = 1, \dots, n \end{array} \right. \end{aligned} \quad (\mathcal{P}_{C_v})$$

$$\begin{aligned} \max \quad & \sum_{A \subseteq N} v(A) d_A \\ \text{s.t.} \quad & \left\{ \begin{array}{l} \sum_{A \subseteq N: i \in A} d_A \leq x_i \quad i = 1, \dots, n \\ d_A \geq 0 \quad \forall A \subseteq N \end{array} \right. \end{aligned} \quad (\mathcal{D}_{C_v})$$

$$\begin{aligned} \max \quad & \sum_{A \subseteq N} v(A) d_A \\ (\Pi'_{C_v}) \quad \text{s.t.} \quad & \left\{ \begin{array}{ll} \sum_{A \subseteq N: i \in A} d_A \leq \sum_{j=1}^m u_{ij} z_{ij} & i = 1, \dots, n \\ \alpha_i \leq \sum_{j=1}^m z_{ij} \leq \beta_i & i = 1, \dots, n \\ \alpha'_j \leq \sum_{i=1}^n z_{ij} \leq \beta'_j & j = 1, \dots, m \\ z_{ij} \in \{0, 1\} \quad \forall i, \forall j \end{array} \right. \end{aligned}$$

As many variables as subsets of agents....

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Towards another linearization of Choquet

Definition To any capacity $v : \mathcal{P}(N) \rightarrow \mathbb{R}$ a mapping $m : \mathcal{P}(N) \rightarrow \mathbb{R}$ called Möbius inverse can be associated, defined by:

$$\forall A \subseteq N, m(A) = \sum_{B \subseteq A} (-1)^{|A \setminus B|} v(B)$$

v can be reconstructed from its Möbius inverse as follows:

$$\forall A \subseteq N, v(A) = \sum_{B \subseteq A} m(B)$$

Definition A capacity is said to be k -additive when its Möbius inverse vanishes for any $A \subseteq N$ such that $|A| > k$, and there exists at least one subset A of exactly k elements such that $m(A) \neq 0$.

- (i) $\forall A \subseteq N, |A| > k \Rightarrow m(A) = 0$
- (ii) $\exists A \subseteq N, |A| = k$ and $m(A) \neq 0$

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Choquet-optimization : linearization 2

$$C_v(x) = \sum_{A \subseteq N} m(A) \min_{i \in A} x_i$$

If Möbius masses $m(A)$ are positive

$$\begin{aligned} \max \quad & \sum_{A \subseteq N} m(A) y_A \\ (\Pi''_{C_v}) \quad \text{s.t.} \quad & \left\{ \begin{array}{ll} y_A \leq \sum_{j=1}^m u_{ij} z_{ij} & \forall A \subseteq N, \forall i \in A \\ \alpha_i \leq \sum_{j=1}^m z_{ij} \leq \beta_i & i = 1, \dots, n \\ \alpha'_j \leq \sum_{i=1}^n z_{ij} \leq \beta'_j & j = 1, \dots, m \\ z_{ij} \in \{0, 1\} \quad \forall i, \forall j \end{array} \right. \end{aligned}$$

- 2-additive case : convexity of v implies non-negativity of all Möbius masses, very compact LP
- 3-additive case : only third-order masses can be negative
- Belief functions : all Möbius masses are positive

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Numerical Tests

with CPLEX 12.1(times in s)

m	$t_{\min}$	m	t_W	t_{C_v}
400	0.36	200	1.29	0.04
800	6.81	300	2.77	1.36
1200	21.25	400	7.51	3.81
1600	56.21	500	18.34	8.79
2000	85.54	600	36.19	18.83
2400	112.27	700	68.39	52.34
2800	181.75	800	120.61	89.37
3200	270.45	900	177.65	165.14
3600	496.93	1000	271.39	342.19

Table 1. Computation times (s) for the paper assignment problem

m : number of items n=m/4: number of agents

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Numerical Tests

with CPLEX 12.1(times in s)

m	$t_{\min}$	m	t_W	t_{C_v}
400	0.56	200	0.49	0.08
800	8.52	300	6.91	3.59
1200	34.15	400	24.42	9.78
1600	95.82	500	77.91	27.75
2000	201.63	600	154.25	55.47
2400	329.29	700	359.54	133.55
2800	550.86	800	518.94	181.91
3200	794.26	900	979.08	393.77
3600	1169.84	1000	1547.87	646.11

Table 2. Computation times for (s) the Santa Claus problem.

m : number of items n=m/4: number of agents

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